

Research Focus and Vision

My research focuses on **Human-AI Interaction in Environmental Design**, integrating Generative AI and Human-Computer Interaction methodologies in two design stages. For the understanding stage, I **diagnose the misalignment** between human and GenAI, and enable targeted **adaptation strategies to enhance GenAI's capability** to perceive and reason, aligning it with human-centric values. For the creation stage, I **transform design intent into computational representations and design interactive systems** that integrate controllable GenAI into design workflows.

Education

- 2022 – now

- Ph.D. Candidate, Computational Media and Arts** at *The Hong Kong University of Science and Technology (Guangzhou)*, supervised by Prof. Wei Zeng and Prof. Kang Zhang.
Thesis: Human-AI Interaction for Environmental Design: Aligning Generative AI with Design Intent from Understanding to Creation.

Human-Computer Interaction

Generative AI
- 2019 – 2022

- M.Eng., Architecture** at *Tongji University*, supervised by Prof. Yu Ye.
Thesis: Economic Benefit Measurement of Street Space Quality and Urban Design Guidance: Based on Multi-source Data and Machine Learning.

Urban Data Analysis

Data-Driven Urban Design
- 2014 – 2019

- B.Arch., Architecture** at *Hunan University, China*

Architecture Design

Urban Design

Publications

I have published 9 peer-reviewed papers at prestigious venues including *ACM CHI*, *IJHCS* and *IEEE TVCG*. **Note about venues:** In the field of the Human-Computer Interaction, the ACM Conference on Human Factors in Computing Systems (*CHI*), and the International Journal of Human-Computer Studies (*IJHCS*) are recognized as the very top tier venues. In the field of Data Visualization, the IEEE Transactions on Visualization and Computer Graphics (*IEEE TVCG*) is recognized as the top-ranked venue.

First Author Publications

- 1

PlantoGraphy: Incorporating iterative design process into generative artificial intelligence for landscape rendering in *Proceedings of ACM CHI Conference on Human Factors in Computing Systems*, 2024.  DOI: 10.1145/3613904.3642824.
[Rong Huang](#), Hai-chuan Lin, Chuanzhang Chen, Kang Zhang, and Wei Zeng* (CCF-A, Citation: 51)
- 2

Synthetic data generation with spatial and semantic fidelity for multimodal large language model on architectural heritage interpretation *Accepted at npj Heritage Science*. 2026.
[Rong Huang](#), Hai-chuan Lin, and Wei Zeng* (SCI Q1)
- 3

Introducing ManyViews: An AI-assisted tool to support citizens' engagement in the design of urban spaces *Accepted at International Journal of Human-Computer Studies*. 2026.
[Rong Huang](#), Yihan Hou, and Wei Zeng* (SCI Q1)

- 4 **SceneWeaver: A multi-agent collaborative system for 3D scene creation in video games** in *Proceedings of the International Symposium on Visual Information Communication and Interaction*, 2025. [DOI: 10.1145/3769534.3769540](#).
[Rong Huang](#), [Chenxi Ruan](#), [Bingchuan Jiang](#), and [Wei Zeng*](#)

Co-Author Publications

- 1 **VISAtlas: An image-based exploration and query system for large visualization collections via neural image embedding** in *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 7, 2024. [DOI: 10.1109/tvcg.2022.3229023](#).
[Yilin Ye](#), [Rong Huang](#), and [Wei Zeng*](#) (SCI Q1, Citation: 40)
- 2 **Is it the end? guidelines for cinematic endings in data videos** in *Proceedings of ACM CHI Conference on Human Factors in Computing Systems*, 2023. [DOI: 10.1145/3544548.3580701](#).
[Xian Xu](#), [Aoyu Wu](#), [Leni Yang](#), [Zheng Wei](#), [Rong Huang](#), [Yip David](#), and [Huamin Qu*](#) (CCF-A, Citation: 15)
- 3 **A data-informed analytical approach to human-scale greenway planning: Integrating multi-sourced urban data with machine learning algorithms** in *Urban Forestry & Urban Greening*, vol. 56, 2020. [DOI: 10.1016/j.ufug.2020.126871](#).
[Ziyi Tang](#), [Yu Ye](#), [Zhidian Jiang](#), [Chaowei Fu](#), [Rong Huang](#), and [Dong Yao](#) (SCI Q1, Citation: 87)
- 4 **Unified visual comparison framework for human and AI paintings using neural embeddings and computational aesthetics** in *IEEE Computer Graphics & Applications*, vol. 45, no. 2, 2025. [DOI: 10.1109/MCG.2025.3555122](#).
[Yilin Ye](#), [Rong Huang](#), [Kang Zhang](#), and [Wei Zeng*](#) (SCI Q3)
- 5 **HeritageExplorer: Interactive visualization and dialogue system for multi-modal architectural heritage exploration** in *Proceedings of the International Symposium on Visual Information Communication and Interaction*, 2025.
[Yusong Wang](#), [Yihan Hou](#), [Rong Huang](#), and [Wei Zeng*](#)

Research Projects

- 2025.01 – 2027.12
- **Guangxi Key Research and Development Program: "Key Technologies for Trustworthy Intelligent Integration of Guangxi Traditional Architecture Based on Multi-modal Data"** (PI: Prof. Wei Zeng)
Funding: ¥600K out of ¥1.5M.
Role: Project Coordinator
 - Contribute to the grant proposal, defining the core technical roadmap and the project timeline.
 - Lead and manage the research team, and responsible for the core technical development and key research outputs.
 - Prepare progress reports and technical documentation.

- 2020.01 – 2022.06 • **National Natural Science Foundation of China:** "Measuring public space quality: An evaluation model and its design support based on multi-sourced urban data and deep learning algorithms" (PI: Prof. Yu Ye)
Role: Student Researcher
- Multi-sourced urban dataset construction using Python and GIS.
 - Statistical analyses to support the development of the evaluation model.
- 2020.01 – 2021.12 • **National Natural Science Foundation of China:** "Walkability of street interfaces: A fine-scale measurement and design control methods" (PI: Prof. Yu Ye)
Role: Student Researcher
- Performed data statistical analysis using Python.
 - Managed the visualization and layout of research findings for publication.

Work Experience

- 2019 – 2022 • **Research Assistant**, the Computational Urban Design Research Centre (PI: Dr. Yu Ye), Joint Laboratory for International Cooperation on Eco-Urban Design (Tongji University), Ministry of Education, China.
- 2021.06 – 2021.09 • **Urban Design Intern**, Skidmore, Owings & Merrill LLP (SOM), Urban Design Department, Shanghai Office, China.
- 2018.06 – 2018.09 • **Architecture Design Intern**, URBANUS architecture and urban design practice, Shenzhen Office, China.

Awards

- 2019.12 • **First Prize**, Shanghai Urban Design Challenge (award ¥100K about \$14,000). The best team among 105 teams from different countries.
- 2016.06 • **Research Prize**, Architecture and Urban Design: Case Study House in USA (award ¥20K about \$3,000). Only the top 1% students were prized among 500 peers.

Skills

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|------------------------|---|
| Design Tools | • Interactive Prototyping (Figma), 3D Modeling & Animation (Rhino, SketchUp, Cinema4D, Unity, Unreal Engine), Adobe Suites |
| Analysis & Programming | • Python, ArcGIS, SPSS, P5.js, PyTorch |
| HCI Methods | • User-Centered Design, Qualitative Methods (Thematic Analysis, Interviews), Quantitative Methods (Statistical Analysis), Prototyping |
| Languages | • Mandarin (Native), English (Professional Proficiency), Cantonese, Hakka |